Cyber Sentinels

Incident Timeline and Evidence Collection for Small Teams

Problem Statement PS-013

Team ID: A3-PS013-TC470

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Abstract and Problem Description

Scattered logs, mixed formats and uncertain clocks make small-team incident reconstruction slow.

Abstract

Cyber Sentinels is a web tool for small teams that investigate security incidents without a dedicated forensic platform. An investigator uploads log files from several hosts and applications. Cyber Sentinels preserves the original bytes, records a SHA-256 hash, parses supported formats into a common event structure, normalises timestamps to UTC while keeping the raw value, and presents one searchable timeline. Records that cannot be parsed are kept and listed as parsing gaps. Reports link each entry back to its source file, line and hash. The expected benefit is a more consistent, reviewable reconstruction with less manual correlation.

Problem description

- **Distributed logs:** events for one incident are spread across hosts and applications, each with its own files.
- **Different formats:** syslog-style text and JSON use different fields, delimiters and severity terms, so correlation by hand is slow and error-prone.
- **Time:** records may use different time zones, omit the year or zone, or come from hosts whose clocks have drifted, so the true order of events is uncertain.
- **Evidence integrity:** original files must stay unchanged and checkable; otherwise later findings are easy to challenge.
- **Manual effort:** searching, cross-checking and writing the report by hand consumes time that small teams rarely have.

Background

A reconstruction is only as trustworthy as its link back to the original evidence.

Reconstructing an incident means placing events from several sources into one sequence and explaining what happened. This needs reliable collection (files received complete and unaltered) [1], a common set of event fields (time, host, source file, line, event type, message) so that records from different formats can be searched together, and a single time reference such as UTC [2]. Raw timestamps must be retained, because a line such as *Oct 14 06:33:01* carries no year or zone and any value assigned to it is an assumption that should be stated.

Normalised records are derived data. Each one must keep a reference to the original evidence (file identifier, hash, line position) so the investigator can return to the raw line. Lines that cannot be parsed should be kept and reported as parsing gaps rather than dropped; otherwise a missing event in the timeline could be mistaken for a missing event in the logs.

Scope and limits. Cyber Sentinels supports an investigator's analysis and reporting; it is not a forensic certification. A matching SHA-256 value shows that a file is identical to the one hashed earlier. It does not, by itself, prove chain of custody, how the file was collected, or that the source system was trustworthy. Clock skew between hosts cannot be corrected without a trusted reference, so timestamps are presented as observed, together with the assumptions made.

Literature & RFC Review

Four sources shape the design; none is claimed as certification or compliance.

Source	What the source specifies	Design decision in Cyber Sentinels (ours)
[1] NIST SP 800-86, Kent et al., 2006	Guidance on using forensic techniques during incident response: collection, examination, analysis and reporting, applied to data from files, operating systems, network traffic and applications. It is guidance, not a legal or step-by-step procedure.	Workflow is organised along the same four stages: upload and preserve (collection), parse and normalise (examination), timeline search (analysis), exports (reporting). The mapping is our choice, not a compliance claim.
[2] RFC 3339, Klyne and Newman, 2002	A date-time format for Internet protocols, a profile of ISO 8601, with an explicit UTC offset, e.g. 2026-10-14T06:33:01Z. (Later updated by RFC 9557.)	Normalised times are stored in UTC and shown in RFC 3339 form; the raw string is kept. The RFC does not say how to treat logs lacking a year or zone, so those assumptions are documented separately.
[3] RFC 5424, Gerhards, 2009	The syslog protocol and message format: priority, version, timestamp, hostname, application name, process ID, message ID, structured data. It obsoletes RFC 3164, the older BSD syslog description.	Its header fields inform the common event fields (host, application, message). Some Linux authentication logs use the traditional timestamp without year or zone, so the parser must state its assumption.
[4] FIPS PUB 180-4, NIST, 2015	Specifies the SHA family of hash algorithms, including SHA-256 (256-bit digest). It does not define evidence handling or chain of custody. NIST has announced a planned revision.	A SHA-256 digest is computed over each uploaded file's original bytes and shown in reports for later integrity checks. A match supports an integrity check only.

The sources define formats and algorithms; how Cyber Sentinels uses them is a design decision. Full references are on the Bibliography page.

Proposed Solution

An eight-step workflow that keeps original evidence separate from derived data.

1. Upload. Investigator submits supported log files through the web interface.	**5. Normalise time.** Timestamps converted to UTC; the raw value and any assumed zone or year are kept.
2. Preserve. Original file bytes are stored unmodified; parsing never alters them.	**6. Retain errors.** Malformed records are kept and listed as parsing gaps.
3. Hash. A SHA-256 digest is calculated and recorded for each file.	**7. Build timeline.** A searchable, filterable timeline of normalised events is generated.
4. Parse. Supported formats are mapped to a common event structure.	**8. Export.** Evidence-linked reports (PDF, CSV, JSON) cite file, line and hash.

Input formats

- **Linux authentication logs** (syslog-style text) and **JSON web logs** are the target formats. They are described here as requirements; their parser implementation is not verified in this document.
- **Synthetic incident datasets** will be used for demonstration and testing, so no real incident data is involved.
- **Time handling:** where a log omits the year or time zone, Cyber Sentinels is designed to apply a user-stated assumption, record it on the event, and never silently invent a missing timestamp.

Optimization Proposed

Expected workflow improvements, not measured results.

Task	Manual workflow	Cyber Sentinels workflow (expected)
Searching across log files	Open or grep each file per host; correlate matches by hand.	One search over normalised events from all uploaded files, with filters.
Evidence integrity checks	Hashes computed ad hoc, if at all; hard to know which copy was analysed.	SHA-256 recorded at upload; original bytes preserved; digest shown in reports.
Time-zone normalisation	Times converted manually; assumptions often unrecorded.	Converted to UTC; raw value and assumption stored with each event.
Handling parsing errors	Unreadable lines are easily skipped or overlooked.	Retained and listed as parsing gaps.
Report preparation	Lines and screenshots copied into a document by hand.	Evidence-linked export in PDF, CSV and JSON.

Triage-time evaluation (planned)

- Give both approaches the **same datasets** (the synthetic incident scenarios) and the **same questions**.
- Fix the **completion criteria** in advance, e.g. identify the first malicious action, the affected account and the last observed event.
- Record time to completion and correctness against a known answer key; alternate which approach is tried first.
- **Measured results: not yet available.** No speed-up or accuracy figures are claimed in this document.

Solution Architecture & Design

Solid outlines: stack named in the project brief. Dashed: proposed, implementation not verified.

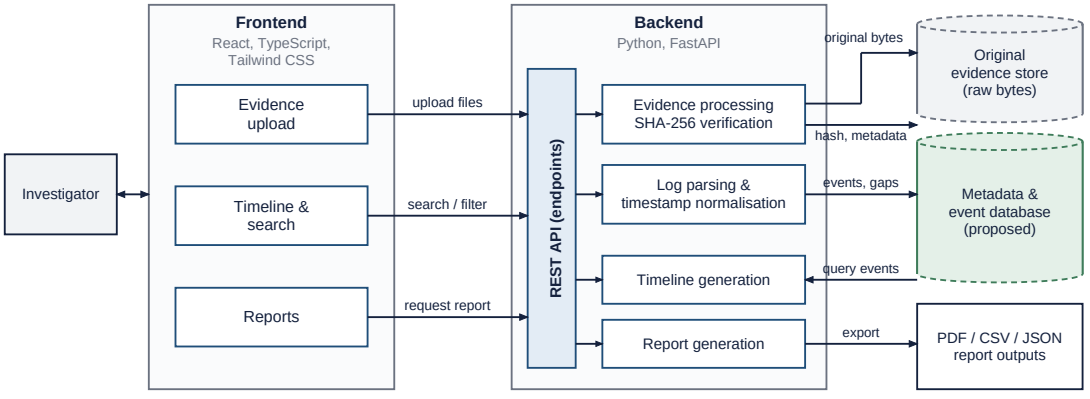


Figure: System Architecture UML Diagram.

Data Flow Diagram

Information flow: original evidence (blue) is kept apart from derived normalised data (green).

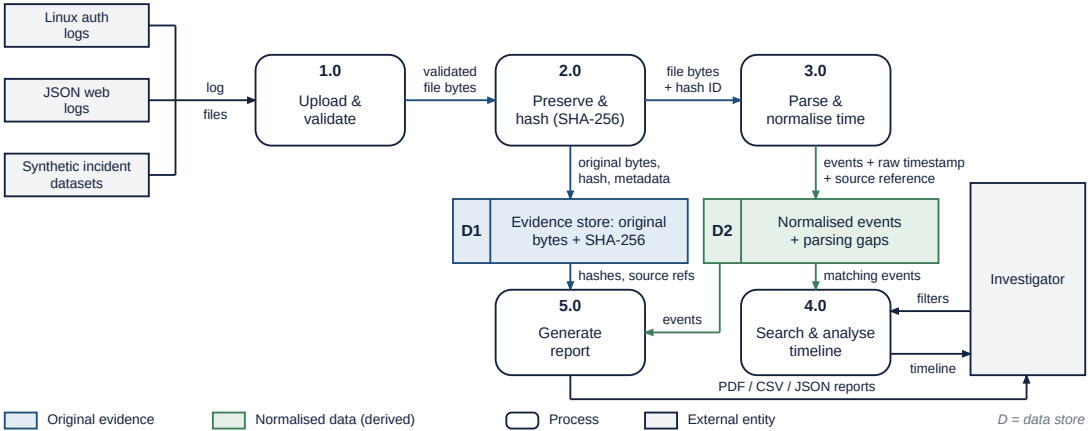


Figure: Data Flow Diagram (DFD).

Deployment Timeline

Five phases from scaffolding to the final demonstration.

Phase	Scope	Status
Phase 1	Project scaffolding and API setup	Completed
Phase 2	Parsers, evidence preservation, hashing and timestamp normalisation	In progress
Phase 3	Timeline interface, search, filters and parsing-gap views	Planned
Phase 4	Three synthetic incident scenarios and automated tests	Planned
Phase 5	Report exports, triage-time comparison, documentation and final demonstration	Planned

Status key: Completed, In progress or Planned. Status reflects the team's own report of current progress; no dates or progress percentages are stated.

Methodology

- Build the evidence path first (preserve, hash, parse, normalise), then the interface on top of it.
- Test parsers with synthetic datasets before any report feature depends on their output.
- Compare manual and Cyber Sentinels triage only after the datasets and completion criteria are fixed.

Bibliography

[1] K. Kent, S. Chevalier, T. Grance and H. Dang, *Guide to Integrating Forensic Techniques into Incident Response*, NIST Special Publication 800-86, Aug. 2006. DOI: https://doi.org/10.6028/NIST.SP.800-86

[2] G. Klyne and C. Newman, *Date and Time on the Internet: Timestamps*, RFC 3339, IETF, July 2002. https://www.rfc-editor.org/info/rfc3339

[3] R. Gerhards, *The Syslog Protocol*, RFC 5424, IETF, Mar. 2009. https://www.rfc-editor.org/info/rfc5424

[4] National Institute of Standards and Technology, *Secure Hash Standard (SHS)*, FIPS PUB 180-4, Aug. 2015. https://csrc.nist.gov/pubs/fips/180-4/upd1/final

[5] AIORI Hackathon 3, *Problem Statement PS-013: Incident Timeline and Evidence Collection for Small Teams*. Official document not available when this report was prepared; add its link here.

[6] Cyber Sentinels project repository, team A3-PS013-TC470. https://github.com/arpit-verma79/A3-PS013-TC470

Reference URLs [1]–[4] were located through web search on 14 October 2026. The repository URL [6] was supplied by the team and was not checked.



Thank You for Your Attention!

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